

Day 10: Review and Practice Practical

Question 1

Open Cisco Packet Tracer and create a new project file named **HomeWiFiNetwork.pkt** with three PCs and one wireless router connected appropriately.

Answer: The Packet Tracer network was created and the topology was connected. The submitted practical screenshot shows the implemented physical topology with a Cisco 1941 router, a 2960 switch and connected PCs. The screenshot contains additional PCs beyond the three requested in the task, but it is the actual topology used for the practical.

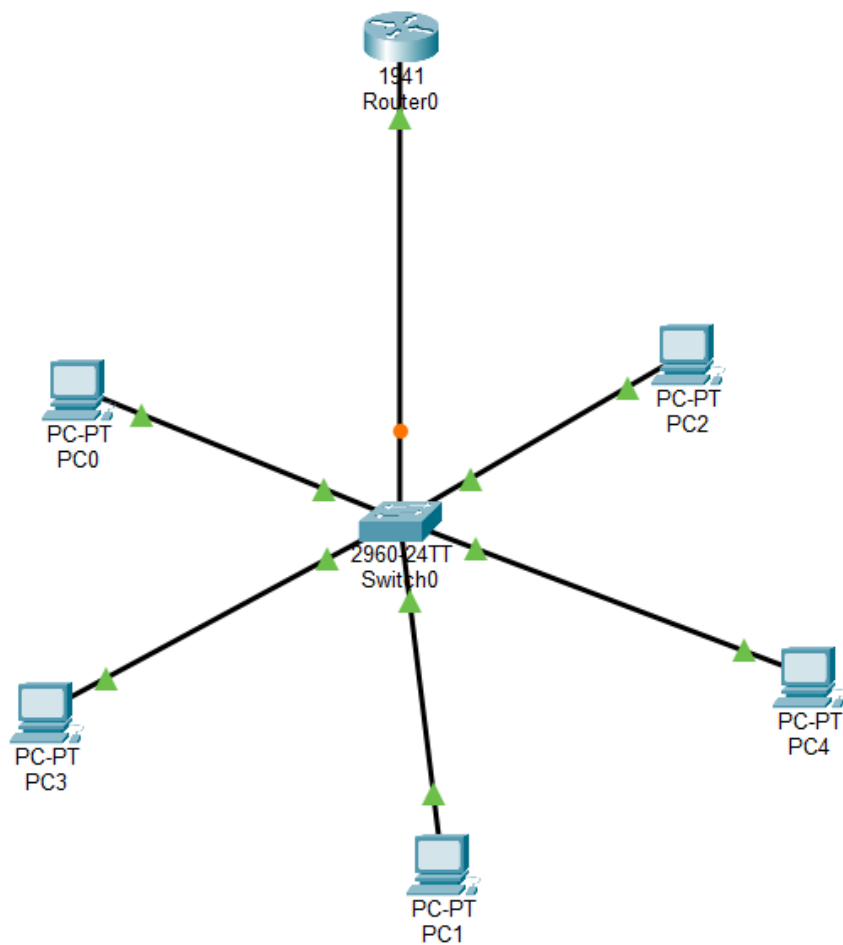


Figure 1: Actual Cisco Packet Tracer topology screenshot.

Question 2

Assign static IP addresses to each PC so they are on the same subnet (for example, 192.168.10.x), and configure the default gateway to the router's IP.

Answer: Static IPv4 addressing was configured using the 192.168.10.0/24 network. The router interface was configured as **192.168.10.1** with subnet mask **255.255.255.0**. PC0 was configured with **192.168.10.2**, subnet mask **255.255.255.0**, and default gateway **192.168.10.1**.

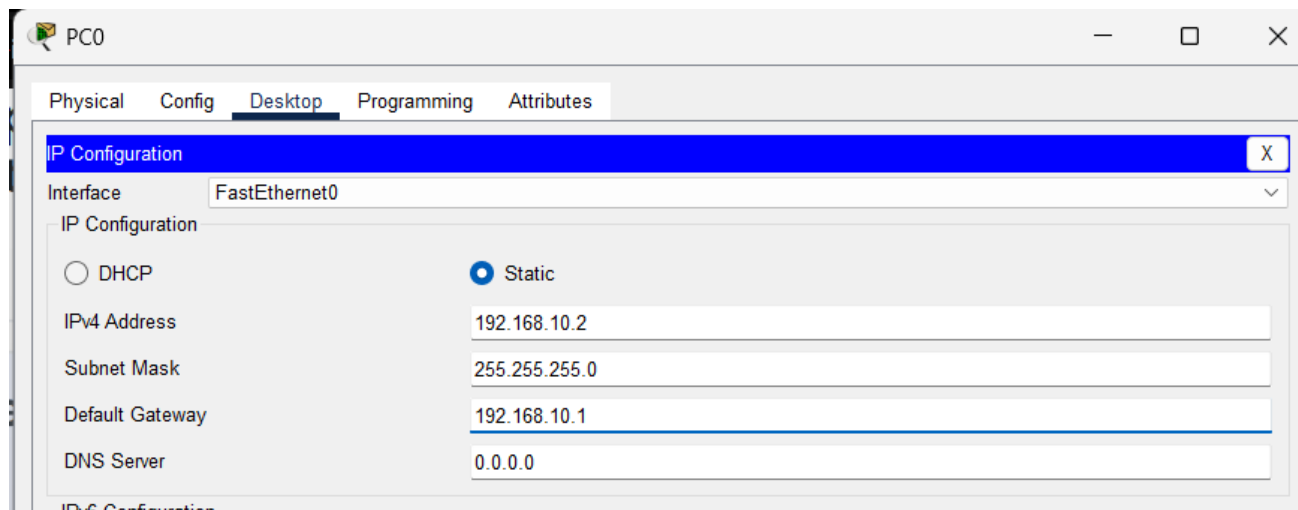


Figure 2: PC0 static IP configuration from the actual Packet Tracer setup.

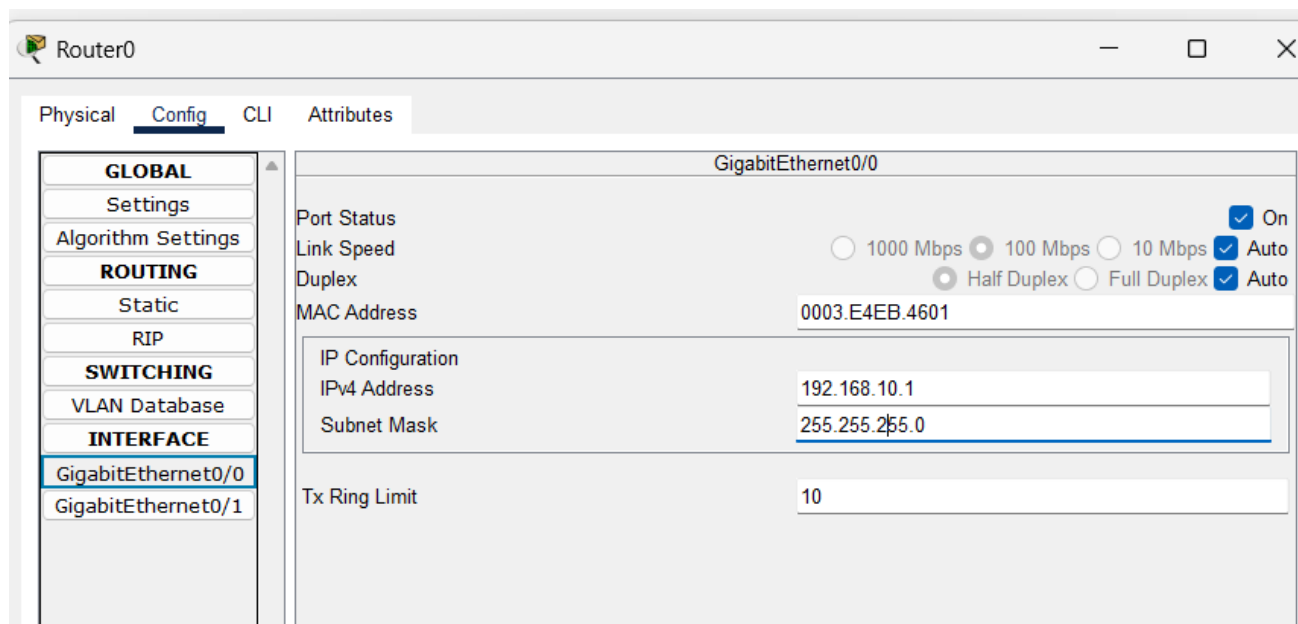


Figure 3: Router GigabitEthernet0/0 configuration showing 192.168.10.1/24.

Question 3

Test connectivity by using the Packet Tracer ping tool to verify that all PCs can communicate with each other and the router. Use Command Prompt and type ping [other PC's IP].

Answer: Connectivity was tested using the Packet Tracer Command Prompt. The captured result shows PC1 successfully pinging **192.168.10.2**. Four replies were received, with **0% packet loss**, confirming successful communication between the PCs.

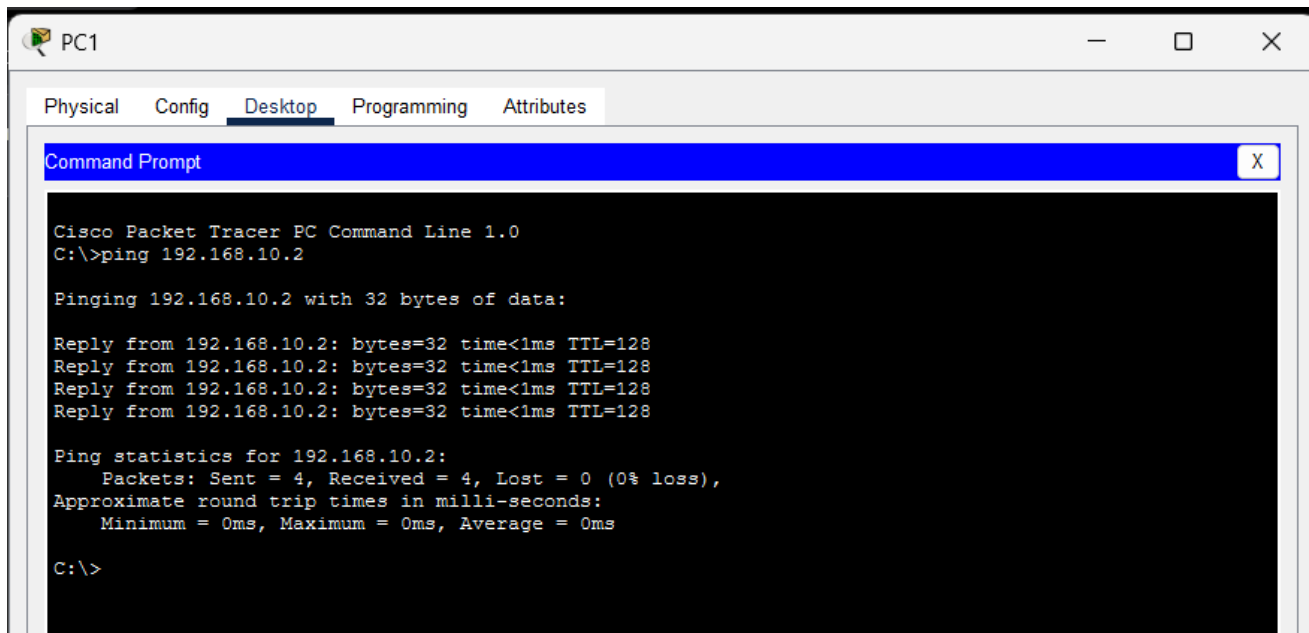


Figure 4: Actual successful ping test from PC1 to 192.168.10.2.

The additional PC2 configuration shown below also uses the same subnet and router gateway, with IP address **192.168.10.6**.

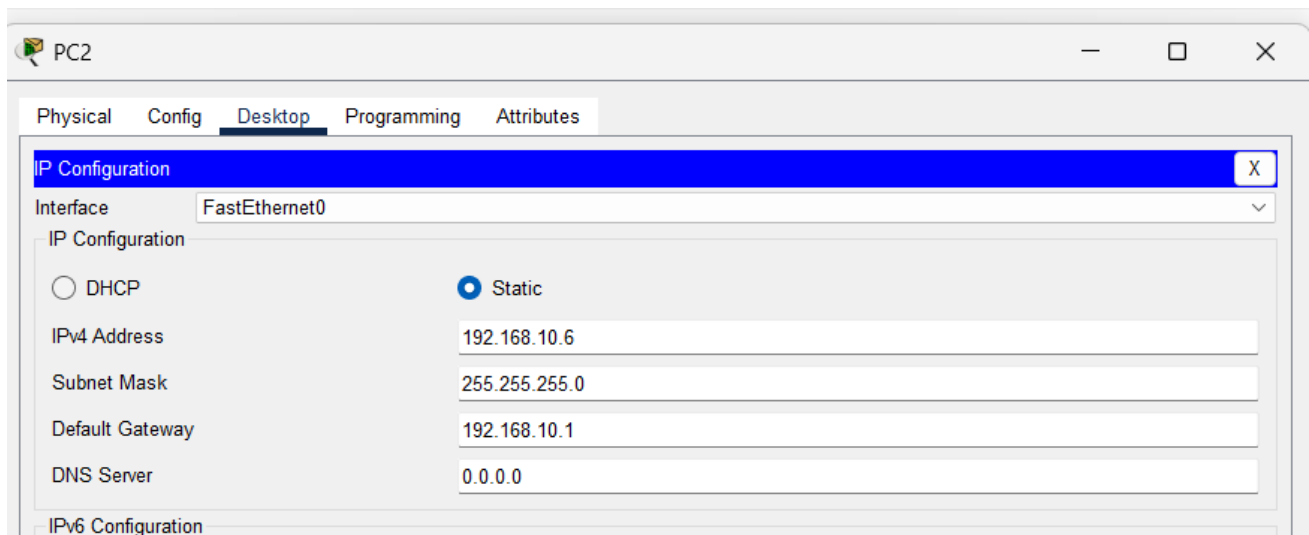


Figure 5: PC2 static IP configuration from the actual Packet Tracer setup.

Question 4

Document each step taken to build and configure the Packet Tracer network, including screenshots of the topology and the IP configuration windows for each PC.

Step 1: Open Cisco Packet Tracer and create a new project. Save the project as **HomeWiFiNetwork.pkt**.

Step 2: Place the router, switch and PCs in the logical workspace and connect them using Ethernet connections.

Step 3: Configure the router's GigabitEthernet0/0 interface with IPv4 address **192.168.10.1** and subnet mask **255.255.255.0**.

Step 4: Configure PC0 with static IP **192.168.10.2**, subnet mask **255.255.255.0**, and default gateway **192.168.10.1**.

Step 5: Configure the other PCs with unique addresses from the same 192.168.10.0/24 subnet and use **192.168.10.1** as the default gateway.

Step 6: Open the Desktop tab on a PC, select Command Prompt, and run **ping 192.168.10.2** or the IP address of another configured PC.

Step 7: Verify the ping result. Four successful replies and 0% packet loss confirm connectivity.

IP Addressing Used

Device	IPv4 Address	Subnet Mask	Default Gateway
Router G0/0	192.168.10.1	255.255.255.0	—
PC0	192.168.10.2	255.255.255.0	192.168.10.1
PC2	192.168.10.6	255.255.255.0	192.168.10.1